

AI Colouring Book

Project Report

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Abstract

Our project revolves around the creation of a local system capable of generating a colouring book suitable for children around the ages of 10, without the need of Api usage. To meet these specific requirements, our team had to conduct a variety of tests to assess which AI models perform efficiently in this environment. A typical colouring book page must contain a picture of the subject/person, which could be a photograph or a portrait. Secondly, a short paragraph under the picture is also necessary to give users an idea of who the person they are colouring is. To better evaluate the best options available, the text and image parts were separated and evaluated individually on the same corpus of 20 preselected people. Four language models were compared based on their capability to produce a fully coherent and informative summary readable by our target audience. Six factual checkers were examined to ensure reliability. On the image side, three pipelines were developed and tested on the same inputs. A user case study was performed on the results of each stage. Two adults rated the text part of the project and reached an agreement of 77.5 percent, with Llama 3.1 8B as the top candidate. On the other hand, the image part was surveyed by five people. Their agreement level was 74.5 percent, with pipeline C as the top performer. Such results in both parts of the project reflect the choices that were followed to create the final product.

1 Introduction

A coloring book of famous people from the University of Marburg is intended to be produced for the university's 500-year celebration. Such a task requires the expertise of several people, including a writer and an illustrator. Maintenance on the book will also require the same amount of effort to change its content. In this project, we aim to research to what extent we can automate this process while working strictly on local hardware. Our system requirements are a 6 GB graphics card, a CPU with a base clock speed of 2.8 GHz, and 16 GB of RAM. What makes this automation challenging is that small local models usually tend to either

omit or add information to the summaries generated. Making it look perfectly fine from an outside perspective but be factually wrong or misleading. The same happens for the clean line art drawings that can be easily coloured but will quietly omit the real person's face features, changing it into a whole different character. In order to test those results, a Wikipedia lead (text and image) was extracted from 20 people. The 20 subjects were randomly selected from different fields and time periods to mimic what the university celebration event might demand. Firstly, the four writers, aka the models assigned to writing the summaries, were given 20 inputs in order to generate 80 short biographies in total. Furthermore, since numeric metrics do not convey basically anything about the text, a human case study was necessary (Sulem et al., 2018). Two adults rated all biographies from each of the four models without knowing which was which. They ranked these summaries from best to worst in terms of three main criteria. The two adults focused on the factual correctness of the text, the difficulty of reading it and the amount of useful information it holds, all while assuming it is meant for children. The order was also shuffled each time to ensure fairness. The same was done for the six checkers, with minor changes in the setup. In this case, 120 claims extracted from the frozen Wikipedia leads were used. The same adults again labelled each claim as "supported", "contradicted" or "Not enough Evidence" (NEI). After comparing the human labels to what the checkers returned, a score was calculated. In the text part, the best writer model was Llama 3.1 8B with a high agreement rate. For the checker, RoBERTa-large MNLI, which is a small sentence-pair classifier, outperformed every other model that had a higher number of parameters. It matched 102 out of the 120 claims, surpassing the second best by 10 matches. In contrast, the image part was tested differently. A group of 5 adults was assembled to pick the best line art out of the three possible pipelines for each person on the list. The drawings were judged based on how complete the lines were, how closely it resembled the original picture, and finally on how easy it is to draw for children. Each

pipeline was covered and annotated by letters and randomised to ensure fairness as well. An automated test was not used due to their unneffectiveness to judge the quality of an image according to our standards (?). Pipeline C performed the best in this test, with the adults having a 74.5 percent agreement rate. The results are summarised in Table 1.

Part	Selected
Writer	Llama 3.1 8B (adult mean rank 1.73 of 4)
Checker	RoBERTa-large MNLI (macro-F1 0.81)
Drawing	Pipeline C, photo-referenced (53 of 100 votes)

Table 1: The three components chosen by the tests in this report.

2 Related Work

During the research phase of this project, several papers and sources were reviewed. In several areas, we drew on these ideas as inspiration for our colorings book system design. Firstly, regarding the text part. The decision was made to instruct the model to extract facts from the Wikipedia lead first, then use simplification and summarization on those facts, rather than doing both tasks at the same time (Cripwell et al., 2023). This fundamentally changes how the model behaves. Furthermore, to adjust the difficulty of the text beyond what the model does already, a dictionary is usually used to change hard, complex words to simplified versions suitable for children. Such a technique posed a huge problem in keeping the meaning of the sentence unchanged (Paetzold and Specia, 2015) and therefore was not used. In contrast, the testing phase used Flesch–Kincaid as a preliminary measure and not as a strict gate due to it only counting words and sentence length, making it not robust for checking if the text was complete or correct (Tanprasert and Kauchak, 2021). Therefore, the user case study involving two adults became mandatory to choose the best candidate. Human verification became necessary even for large models that have a way lower chance of hallucinating (Kew et al., 2023). On the other hand, the checkers are necessary due to the writer posing a huge factuality problem. The writer, when generating a summary or while simplifying, can introduce new additions to the sentences not found in the source(lead) (Maynez et al., 2020; Devaraj et al., 2022). The checker therefore uses the sentence as a whole and does not divide it into separate claims, in the sense that it checks if the

whole sentence is correct rather than keeping what is true and just omitting the false additional claim (Min et al., 2023). For example, the sentence here, “Einstein became an American citizen in 1935”, is a false addition that keeps the event of Einstein becoming an American citizen but adds a wrong date to it, making it completely false. Therefore, in such a test, two NLI models were also added due to their specific utility. These small models perform better when the source is already cut into short pieces (Laban et al., 2022). The system does that by comparing each biography sentence with pairs of sentences from the lead and generating a label. Secondly, the image part has relied on informative drawing to ensure that the line drawings keep all the shapes and structures as close to what people saw in the real picture (Chan et al., 2022). That alone would not suffice to keep the drawing as faithful to the source as possible. Therefore, the IP-Adapter allows Stable Diffusion to check the reference picture while redrawing the output of ID (Ye et al., 2023) (Rombach et al., 2022). In addition to that, a ControlNet is also attached to SD to guarantee that it follows the lines of said ID (Zhang et al., 2023). The combination of all these components is what our research led us to. Finally, the resources found on the internet were all focused on different scenarios not mirroring our situation. Our system is evaluated on a single path from a name on Wikipedia directly to a printable A4 page. Each part: writer, checker and the drawing have to pass a human check to be ready for deployment. Faithfulness to the source is therefore a necessity rather than just a metric in our research.

3 Data and Resources

During this research, a collection of biographies of 20 people from Wikipedia was accumulated. The headings were removed, and only the opening section was kept while omitting any text in parentheses. These 20 subjects were diverse and contained scientists, artists, athletes and so on. To have a fixed corpus, the leads were downloaded on 16 September 2026 and frozen with a hash on the local computer. Each lead was recovered in English from Wikipedia’s public interface. For contrast, the checker part used 60 claims from the four writers, and another 60 claims that were constructed as 20 true, 20 false, and 20 not decidable by Grok 4.7 to ensure that the dataset is diverse. In total, the dataset contains 80 biographies made by the

four writers, 60 drawings from the three pipelines and 120 checker claims. In terms of labelling, two adults ranked all generated biographies from best to worst. In addition to that, they labelled all 120 claims as SUPPORTED, CONTRADICTED, or NEI based on what’s found in the lead. In contrast, five people picked the best drawing for each subject on the list. Therefore, all data aside from the lead were labelled by humans to avoid model hallucination.

4 Method

The system is built to retrieve a frozen lead that was previously downloaded from Wikipedia from a string (name) input. The system then generates a biography simplified for children, accompanied by a drawing and a name for that input.

4.1 Writer and Checker

The text generated is checked, and if all statements are supported, the result is an A4 printable page ready to be added to the coloring book. The system is built using local models that have to be downloaded before use. Four instruct-tuned models were considered and evaluated. These were: Qwen 2.5 7B, Llama 3.1 8B, Llama 3.2 3B, and Mistral 7B. The models were chosen based on how demanding they were in terms of graphics card requirements. The local computer was able to handle up to 8 billion parameters with 6gb of VRAM due to Ollama allowing compression under the "Q4_K_M" format. Therefore, each writer given a Wikipedia lead was instructed to list all important facts and then rewrite them for a 10-year-old child (Cripwell et al., 2023). Moreover, the writers had to stick to an upper limit of six sentences with 14 words each, while keeping all important dates, names and numbers. However, the design choice behind the number of sentences and words was purely a layout choice to make the final printable PDF look organized. Hence, it must not be considered a proven limit on what children could understand and should be treated as such. The baseline here is that the four writers are compared with each other while the readability grade is reported only as background, not as the score that picks the winner. In contrast, six checkers were examined throughout the course of this research. Four of them are the same models used for the writers. Furthermore, letting a writer judge their own text was an intentional addition to show the effect of bias (Panickssery et al., 2024). Each judge labels

every sentence either “supported” if all facts in it exist in the lead, “Contradicted” if at least one fact contradicts the lead, or “NEI” if at least one fact is not in the lead, meaning it’s not either supported or contradicted. Moreover, a fact can be true, but if it does not appear in the frozen lead, it is also treated as “NEI”. To support such labels, each model must return the label plus the number of the sentences it uses as proof of its decision-making. On the other hand, the two remaining checkers are DeBERTa-v3-base and RoBERTa-large MNLI. These models act differently and check each sentence from the generated biographies against every chunk of the lead, not the full thing at the same time (Laban et al., 2022). Each chunk then gets a support and contradiction score and is kept in memory. After the full run, the highest support and contradiction scores are registered and returned with a minimum score cut-off of 0.8.

4.2 Drawing

The baseline here is labelling all claims as Supported, since a model such as that would look strong on clean text. Therefore, our selected checker must beat it. Once the biographies are checked by these six checkers, the portrait side is handled on its own. The image pipeline receives the picture from the Wikipedia lead and feeds it directly to the Informative Drawings model (Chan et al., 2022). The model weights were previously downloaded from Hugging Face with the name `awacke1/Image-to-Line-Drawings`. The output is then given to Stable Diffusion 1.5 with the name of the person in the portrait and a prompt that asks for a recognisable face, clean black outlines, plain clothes, little interior detail, and large white areas (Rombach et al., 2022). while the negative prompt bans shading, photo texture, dense patterns, and background objects. The mentioned setup is considered pipeline A and the baseline of the experiment. Pipeline B then adds a ControlNet to that same redraw to align Stable Diffusion with the outlines of the informative drawing model sketch (Zhang et al., 2023). Pipeline C then builds on that and incorporates an IP-Adapter, which allows SD to access the original portrait from Wikipedia as a reference, further guiding into the right direction (Ye et al., 2023). All pipelines therefore share the same prompt, size (512×640), and input. While using the UniPC sampler for 40 steps, with an image-to-image strength of 0.57 and a guidance scale of 9. In pipelines B and C, the ControlNet has

a scale of 0.90 for the first 95 percent of the steps, and for the IP-Adapter in Pipeline C, its scale was 0.35. Finally, the last layer before the final drawing is printed, a post-processing step is applied. The drawing is turned grey, and then every pixel darker than 205 becomes black, and every lighter pixel becomes white in order to make it more suitable for coloring.

Pipeline	Stable Diffusion redraw	ControlNet	IP-Adapter
A	yes	no	no
B	yes	yes	no
C	yes	yes	yes, sees the original photo

Table 2: Pipeline A only redraws the line drawing. Pipeline B adds ControlNet. Pipeline C adds IP-Adapter as well, so the generator also sees the original portrait.

5 Evaluation

In order to evaluate the system, each component was assessed separately. Two adults rated the four biographies generated for each input by the models. The instruction was to rank these biographies based on how informative and truthful they are, in addition to their reading difficulty. Therefore, the score is the mean rank over 40. The results are shown in Table 4, and a snippet of the survey is shown in Table 3. For the checker, each label produced is compared against human labels on 120 claims. The two adults rated the claims without knowing the identity of the model or whether it was constructed or real. They gave the same label 99.2% of the time. To keep the weight of rare labels such as Contradicted and NEI the same as the supported claims, the macro F-1 score was used. In addition to that, recall is also calculated based on how many human-labelled claims the checker found correctly for each label. In contrast, if the language-model checker does not return one of the three valid labels accompanied by the numbers of the Wikipedia sentences it used, it is automatically counted as invalid, does not get guessed and is counted as wrong. The image part was evaluated differently. Five people were tasked to be judges for the three pipelines. Each pipeline produced 20 drawings corresponding to the 20 people list. For every input, the judges had to pick the best drawing based on three main criteria. Firstly, the completeness of the lines. Secondly, the resemblance of the output to the original picture, and finally, the colouring difficulty of the sketch. The order of each pipeline was randomised

at each input to ensure fairness (van der Lee et al., 2019). The score in this evaluation was computed by the number of times a pipeline was chosen over 100. To conclude the evaluation, it is also important to mention that no significance test was performed.

Letter	Biography shown to the rater	Rater 1	Rater 2
A	Albert Einstein was a scientist from Germany. He was born on 14 March 1879. He won a big prize in 1921 for his work. He wrote four important papers in 1905. He came up with a new way to understand gravity in 1915. He had to leave Germany in 1933 because of bad people.	4	4
B	Albert Einstein was born in 1879. He died in 1955. He made the theory of relativity. He won a prize for his discovery. He wrote important papers in 1905. He proposed general relativity in 1915.	2	2
C	Albert Einstein was a German-born science expert who lived a long time. He is famous for his big idea about time and space. He wrote a famous mathematical rule: $E = mc^2$. He received a big prize in 1921 for his discovery. Einstein moved to Switzerland in 1895 and stayed there. He warned President Roosevelt about a big danger.	3	1
D	Albert Einstein was born in the German Empire on March 14, 1879. He moved to Switzerland in 1895, where he stayed for the rest of his life. He worked at a patent office. In 1905, he earned a PhD from the University of Zurich. That year, he wrote four significant papers, one of them about the photoelectric effect.	1	3

Table 3: One writer sheet, Albert Einstein. Rank 1 is the best. The model name was hidden behind the letters.

6 Results and Discussion

6.1 Writer

The results of the evaluation are presented in Table 4. The adults ranked Llama 3.1 8B as the best writer with a mean rank of 1.73, while taking first place in 19 out of 40 rankings. Mistral 7B held up second place with a mean rank of 2.33, showing a significant gap between the two. On the other hand, Qwen ranked the lowest in both mean rank and readability, making the worst-performing model in the human user study, proving that easy reading is not the same as a good biography (Tanprasert and Kauchak, 2021). An example is shown in Figure 1,

where it is visible that Qwen changed Einstein’s Nobel Prize to “He won a prize for his work”. Therefore, changing the meaning and significance of such an achievement.

Writer	Mean rank	Ranked first
Llama 3.1 8B	1.73	19/40
Mistral 7B	2.33	12/40
Llama 3.2 3B	2.38	8/40
Qwen 2.5 7B	3.58	1/40

Table 4: Two adults ranked the four biographies of each person (20 people $\times$ 2 raters). Lower mean rank is better.

Best-ranked writer (Llama 3.1 8B), Albert Einstein.
 Albert Einstein was born in the Kingdom of Württemberg in 1879. He made a famous theory about how the universe works. He wrote a famous math formula called $E=mc^2$. He received a big award in 1921 for his work in physics. He moved to Switzerland in 1895 and became a Swiss citizen. He told President Roosevelt about a danger from Germany.

Worst-ranked writer (Qwen 2.5 7B), Albert Einstein.
 Albert Einstein was born in 1879. He died in 1955. He made theories about relativity. He won a prize for his work. He moved to Switzerland in 1895. He wrote important papers in 1905.

Figure 1: The best- and worst-ranked biographies of the same person, as shown to the raters. Both are short and easy to read. Although the second one drops the details that make the facts specific.

In contrast, the same test was handed over to an external llm. Grok 4.7 ranked these generated biographies as shown in Table 5. Llama 3.1 8B still came on top with a mean rank of 2.3. Although Llama 3.2 3B outperformed mistral in this evaluation, Qwen kept its last spot with a mean rank of 2.7. Grok had all the models close in ranking with the highest difference at 0.3.

Writer	Mean rank	Ranked first
Llama 3.1 8B	2.30	5/20
Llama 3.2 3B	2.35	7/20
Mistral 7B	2.65	5/20
Qwen 2.5 7B	2.70	3/20

Table 5: The same ranking task done by a non-candidate language model. It supports, but does not replace, the adult ranking.

6.2 Checker

The final labels provided by the adults in the user case study resulted in a total of 78 SUPPORTED,

20 CONTRADICTED and 22 NEI. Table 6 compares all six checkers with the baseline.

Checker	Macro-F1	C	NEI	S
RoBERTa-large MNLI	0.81	1.00	0.77	0.83
DeBERTa-v3-base	0.59	1.00	0.50	0.55
Mistral 7B	0.57	0.05	0.82	0.94
Qwen 2.5 7B	0.56	0.05	0.91	0.87
Llama 3.1 8B	0.46	0.00	0.68	0.83
Llama 3.2 3B	0.23	0.15	0.00	0.30
Always SUPPORTED	0.26	0.00	0.00	1.00

Table 6: Checker scores on 120 human-labelled claims. Macro-F1 weighs all three labels equally. C, NEI, and S are the share of each label the checker found.

RoBERTa-large MNLI was the clear winner with a macro-F1 of 0.81. Furthermore, resampling was applied by picking 20 people in different mixes, resulting in RoBERTa’s score staying between 0.75 and 0.87. The score remains above the suggested baseline and validates the model as the clear choice. In terms of raw numbers, the latter found all 20 contradicted claims and 17 of the 22 NEI claims while missing 13 of the 78 supported claims. Additionally, DeBERTa-v3-base is second with a macro-F1 0.59. On the other hand, all language models rarely caught a contradiction. Mistral, even though it matched more claims than second place by achieving 92 of 120, only found 1 out of 20 contradictions. Its macro score therefore dropped to 0.57. In addition to that, Qwen also only found 1 of 20 contradictions while Llama 3.1 found none. Llama 3.2, which is the smallest llm out of the bunch, performed below the baseline with a score of 0.23 due to the majority of its output being invalid. Therefore, a larger model does not constitute a better fact checker than a small classifier trained on the same task required.

6.3 Drawing

The five adults cast their votes for their favourite drawing across three pipelines. 100 votes have been gathered for 60 drawings. Table 7 displays the results. A drawing is chosen as a majority winner if it has at least 3 out of 5 votes.

Pipeline	Votes	Majority for
A: SD image-to-image	36/100	8/20 people
B: A + ControlNet	11/100	1/20 people
C: B + IP-Adapter	53/100	11/20 people

Table 7: Adult preferences for the three image pipelines.

Pipeline C gathered 53% of the votes, with a majority for 11 people from the list. Seven of which were unanimous, meaning all five adults agreed on the same drawing. Notable names include Alan Turing, Nelson Mandela, and Frida Kahlo. Therefore, these results suggest that adding a reference image to the real input assists the line-based conditioning applied by the ControlNet to create appealing sketches. However, such benefits were not consistent all throughout. Pipeline A without a ControlNet and an IP-Adapter managed to receive 36 of 100 votes and a majority for 8 people. Marie Curie, Jane Austen, and Hayao Miyazaki were all unanimous votes, making the pipeline a clear favorite for these inputs. Hence, the additional conditioning layers do not always provide a better illustration or sketch. On the last place, pipeline B received 11 of 100 votes and the majority for 1 person. Showcasing that ControlNets do not significantly improve upon the simpler image-to-image baseline. To summarise these results, pipeline C comes out on top as our main choice for the system. However, the judges picked one favourite sketch only. They did not give a separate score for line completeness, resemblance, and colouring difficulty, so the votes do not show which of those made C win. The comparison used only one seed per person and five adult participants. The 100 votes are repeated judgements of the same twenty subjects, not 100 independent image samples. Figure 2 shows the variation between the best and worst outputs from the same system.



Figure 2: A weak drawing (left) and a strong drawing (right) from the same system. On the left the face breaks apart and the hair becomes loose scribbles. On the right the outlines are closed and the person is recognisable.

6.4 Errors

Writer models are prone to hallucinations and can make two kinds of errors. They either add facts that do not exist in the lead to the generated text. Such additions can be small life details such as a favourite hobby or a pet’s name. On the other hand,

the model could change the meaning of a sentence by removing a specific detail that held up the claim. The Einstein example perfectly showcases such an error. The first kind is caught by checker models, while the second is only handled by humans due to intricacies in the context. Language model checkers also fail in two distinct ways. Statements that sound plausible are still accepted even though they are false due to their inability to always distinguish facts. The second problem comes from the model’s failure to comply with an output format, making it invalid and unreadable. The NLI models, on the other hand, struggle with some paraphrased sentences and reject them when the wording is very different. Such an error was permissible due to the system being designed to be rather overprotective of truthfulness. Finally, the drawings also fail in some scenarios. When the input has complicated detail, the lines break. The model adds new interior detail, as seen in Figure 2, where the face falls apart, and the hair turns into loose scribbles. Moreover, faint lines also disappear in the black-and-white post-processing step. That is why the human vote captures the overall impression but does not say which of these happened.

6.5 Limitations

To clarify our work, a sample of 20 people/inputs is not sufficient to draw conclusions on what model or pipeline is best in general. Therefore, our claims only hold for this sample. In addition to that, every human judge has been an adult even though the research is presented as a colouring book for children. We could not get the required documentation and permission to properly conduct a case study on children of the age 10-12. Furthermore, the number of adult human judges is also not sufficient to prove any of our claims beyond this group. Our evaluation sheets and metrics require a 2-hour effort from the human contributors, making it difficult to distribute over a larger sample. Moreover, the drawing vote is one favourite per person and not a separate score for likeness, complete lines, and ease of colouring, making a one-dimensional ranking. Therefore, a later test would be conducted in the local region of Marburg using the real target demographic in order to verify whether the children would understand and enjoy coloring the pages.

6.6 Final result

Our final results can be seen as follows after running the system on a selected name. For this result

we selected Mahatma Gandhi (Figure 3).

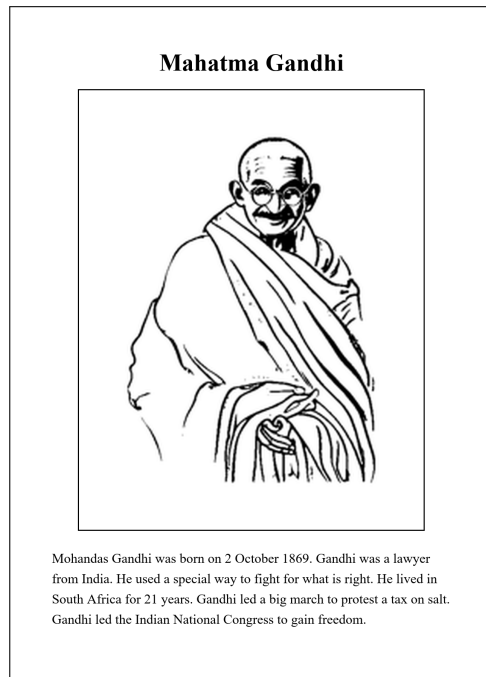


Figure 3: The finished A4 page for Mahatma Gandhi: the name, the pipeline C drawing, and the Llama 3.1 8B biography.

7 Conclusion

This project presents an automation workflow for generating colouring-book pages from Wikipedia text and portraits using local models. In this sample adults preferred Llama 3.1 8B, while RoBERTa-large MNLI scored the highest macro-F1 on the 120 claims. Therefore, these two models were selected, respectively, as the best writer-checker pair. Pipeline C became the main method used for drawings. These results support component selection within the tested setting rather than establishing universal superiority. Future work would include a user case study on children as mentioned in the limitations, to further inspect the quality of the system and improve it accordingly.

8 Declaration of Generative AI Use

Throughout the course of the project, external generative models were used as supplementary tools. Examples are ChatGPT and Grok. The primary function was to help with research material to efficiently allocate our time. In addition to that, LaTeX formatting was guided by AI to ensure the structure of the paper. Citations were also generated by the same helping models for the same reason. For the evaluation part, the sheets that were handed

to the human judges were all generated by AI for simplicity and time constraints. Finally, the code documentation and the prompts for all the local models were assisted by LLM’s and paraphrased by us.

9 Contribution Statement

Name	Contribution
Wilson	Implemented the pipeline A using Awacke and stable diffusion 1.5 and pipeline C using 2 additional components, ControlNet and IpAdapter. Conducted image test for humans to evaluate. Also, wrote the image-method and result section of the report.
Houssam	Designed and programmed the biography-generation and fact-checking pipelines, and validated them through writer and checker tests for human evaluation. Subsequently wrote the test-method and results section of the report.
Melroy	Impelemented pipeline B using awacke, stable diffusion, ContolNet. Wrote the Abstrat, Introduction, Conclusion, Delectraion of Genertive Ai use, Refernces and added citations.

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